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<?xml version="1.0" encoding="utf-8"?>
<?xml-stylesheet type='text/xsl' href='../mesmer1.xsl'?>
<me:mesmer xmlns="http://www.xml-cml.org/schema"
xmlns:me="http://www.chem.leeds.ac.uk/mesmer"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
<title> Glyoxal</title>
<moleculeList>
  <molecule id="N2">
    <atomArray>
      <atom id="a1" elementType="N" />
      <atom id="a2" elementType="N" />
    </atomArray>
    <bondArray>
      <bond atomRefs2="a2 a1" order="3" />
    </bondArray>
    <propertyList>
      <property dictRef="me:epsilon">
        <scalar>82</scalar>
      </property>
      <property dictRef="me:sigma">
        <scalar>3.74</scalar>
      </property>
      <property dictRef="me:MW">
        <scalar units="amu">28.0</scalar>
      </property>
    </propertyList>
  </molecule>
  <molecule spinMultiplicity="2" id="OH">
    <atomArray>
      <atom id="a1" elementType="O" x3="0.000000" y3="0.000000"
z3="0.107992"/>
      <atom id="a2" elementType="H" x3="0.000000" y3="0.000000" z3="-
0.863936"/>
    </atomArray>
    <bondArray>
      <bond atomRefs2="a2 a1" order="1"/>
    </bondArray>
    <propertyList>
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        <scalar units="kJ/mol">0.0</scalar>
      </property>
      <property dictRef="me:rotConsts">
        <array units="cm-1">19.2438</array>
      </property>
      <property dictRef="me:symmetryNumber">
        <scalar>1</scalar>
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      <property dictRef="me:vibFreqs">
        <array units="cm-1">3722.0</array>
      </property>
    </propertyList>
  </molecule>
</moleculeList>
</me:mesmer>

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    <property dictRef="me:MW">
      <scalar units="amu">17</scalar>
    </property>
    <property dictRef="me:spinMultiplicity">
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    </property>
    <property dictRef="me:electronicExcitation">
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    <property dictRef="me:frequenciesScaleFactor" default="true">
      <scalar>1</scalar>
    </property>
  </propertyList>
  <me:DOSCMMethod default="true" name="QMRotors" />
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    <atom id="a1" elementType="C" x3="-0.322384" y3="0.273379"
z3="0.417795"/>
    <atom id="a2" elementType="O" x3="0.221696" y3="1.157088" z3="-
0.213192"/>
    <atom id="a3" elementType="O" x3="-1.251652" y3="-0.586042" z3="-
0.082974"/>
    <atom id="a4" elementType="H" x3="-0.138334" y3="0.045272"
z3="1.484393"/>
    <atom id="a5" elementType="H" x3="-1.376187" y3="-0.343589" z3="-
1.018654"/>
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  <bondArray>
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    <bond atomRefs2="a2 a1" order="2"/>
    <bond atomRefs2="a3 a1" order="1" id="b1"/>
    <bond atomRefs2="a1 a4" order="1"/>
  </bondArray>
  <propertyList>
    <property dictRef="me:ZPE">
      <scalar units="kJ/mol">0</scalar>
    </property>
    <property dictRef="me:vibFreqs">
      <array units="cm-1"></array>
    </property>
    <property dictRef="me:spinMultiplicity">
      <scalar units="cm-1">1</scalar>
    </property>
    <property dictRef="me:hessian">
      <matrix matrixType="squareSymmetricLT" rows="15" units="Hartree/Bohr2">
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</propertyList>
<me:DOSCMMethod default="true" name="QMRotors" />
<me:ExtraDOSCMMethod xsi:type="me:HinderedRotorQM1D">
  <bondRef>b1</bondRef>
  <me:HinderedRotorPotential format="numerical" units="kJ/mol"
expansionSize="10" UseSineTerms="yes">
    <me:PotentialPoint angle=" 0 "potential=" 0 "/>
    <me:PotentialPoint angle=" 30 "potential=" 12.44487 "/>
    <me:PotentialPoint angle=" 60 "potential=" 38.8574 "/>
    <me:PotentialPoint angle=" 90 "potential=" 55.319285 "/>
    <me:PotentialPoint angle=" 120 "potential=" 47.285255 "/>
    <me:PotentialPoint angle=" 150 "potential=" 28.40791 "/>
    <me:PotentialPoint angle=" 180 "potential=" 19.559975 "/>
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    <me:PotentialPoint angle=" 240 "potential=" 47.285255 "/>
    <me:PotentialPoint angle=" 270 "potential=" 55.319285 "/>
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    <me:PotentialPoint angle=" 330 "potential=" 12.44487 "/>
    <me:PotentialPoint angle=" 360 "potential=" 0 "/>
  </me:HinderedRotorPotential>
  <me:CalculateInternalRotorInertia phaseDifference ="180.0"/>
</me:ExtraDOSCMMethod>
</molecule>
<molecule spinMultiplicity="2" id="Pre1">
  <atomArray>
    <atom id="a1" elementType="C" x3="1.092810" y3="0.175496"
z3="0.020738"/>
    <atom id="a2" elementType="O" x3="0.355551" y3="1.136637"
z3="0.001187"/>
    <atom id="a3" elementType="O" x3="0.706523" y3="-1.096977" z3="-
0.028274"/>
    <atom id="a4" elementType="H" x3="2.183878" y3="0.246463"
z3="0.080632"/>
    <atom id="a5" elementType="H" x3="-0.272242" y3="-1.101711" z3="-
0.079840"/>
    <atom id="a6" elementType="O" x3="-1.886176" y3="-0.147812"
z3="0.068940"/>
    <atom id="a7" elementType="H" x3="-1.875688" y3="0.667496" z3="-
0.460040"/>
  </atomArray>
  <bondArray>
    <bond atomRefs2="a7 a6" order="1"/>
    <bond atomRefs2="a5 a3" order="1"/>
    <bond atomRefs2="a3 a1" order="1"/>
    <bond atomRefs2="a2 a1" order="2"/>
    <bond atomRefs2="a1 a4" order="1"/>
    <bond atomRefs2="a5 a6" order="1"/>
  </bondArray>
</propertyList>
<property dictRef="me:ZPE">

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    <scalar units="kJ/mol">-22.84574138</scalar>
  </property>
  <property dictRef="me:vibFreqs">
    <array units="cm-1"></array>
  </property>
  <property dictRef="me:spinMultiplicity">
    <scalar units="cm-1">2</scalar>
  </property>
  <property dictRef="me:hessian">
    <matrix matrixType="squareSymmetricLT" rows="21" units="Hartree/Bohr2">
      0.63514756 0.10206277 0.23447383 0.04510834 -0.10133634 0.97740156 -
      0.16081559 0.00148969 -0.16051799 0.17495275 -0.00041655 -0.07913543
      0.04508925 0.00389904 0.03766745 -0.14145932 0.04957171 -0.65896285
      0.22525464 -0.04540028 0.73063533 -0.29714226 -0.05208773 -0.04218471 -
      0.03191326 -0.00517189 -0.07241505 0.51577471 -0.05413598 -0.07592192
      0.00047244 -0.00430353 0.02892632 -0.01049748 0.12545890 0.08720530 -
      0.02489535 0.00448542 -0.13771969 -0.07432692 -0.01008382 -0.02443379 -
      0.06904104 -0.07846524 0.48154481 -0.14226409 -0.03535148 0.11393944
      0.01602912 0.00065607 -0.01392798 -0.03303739 -0.01434041 0.02461310
      0.16409166 -0.03690310 -0.08036158 0.04663281 -0.00286312 0.01927359
      0.00523490 -0.00933155 0.00756675 0.00636115 0.05135378 0.04735158
      0.12307486 0.04891254 -0.19474548 0.01058227 0.01131854 -0.04004271 -
      0.00855807 -0.00178971 0.00864870 -0.12168543 -0.05836134 0.22477289 -
      0.04243875 -0.01693190 0.03807336 0.00727822 0.00233318 0.00338614 -
      0.14967110 -0.05097564 0.14226151 -0.00442861 -0.00234169 -0.00265199
      0.19137752 -0.01220287 0.00065041 0.00644694 0.00373617 -0.00563628
      0.00149519 -0.05651514 -0.04918124 0.07617208 -0.00219729 0.00608387
      0.00025169 0.06647960 0.05102427 0.01014964 -0.00048985 0.02711509 -
      0.00422366 -0.00111961 -0.01391783 0.17700230 0.08424789 -0.31414507 -
      0.00418019 0.00015065 0.00031988 -0.17656786 -0.08129444 0.29501081
      0.00684603 0.00055575 0.00404039 0.00008550 0.00012236 -0.00208535 -
      0.00240098 -0.00059254 0.00080241 -0.00049930 0.00005751 -0.00068052 -
      0.00332531 -0.00093895 -0.00022551 0.31584635 0.00208526 -0.00017155
      0.00210907 -0.00236907 0.00105049 -0.00041386 -0.00290692 0.00213780
      0.00370128 -0.00022328 -0.00041683 -0.00021030 0.00154634 -0.00471021 -
      0.00427695 -0.17880551 0.10719325 -0.01217041 -0.00196317 -0.00925235
      0.00230440 0.00068946 0.00588572 0.01356969 0.00608598 -0.01277459
      0.00124222 -0.00012562 0.00065223 -0.00181222 -0.00443964 0.00462458
      0.16542732 -0.09307392 0.09992261 0.00066709 0.00026291 0.00154117 -
      0.00561674 -0.00142221 0.00124692 -0.00160972 -0.00111080 0.00058630
      0.00010861 0.00002816 -0.00008111 0.00120802 0.00163849 -0.00195471 -
      0.31655228 0.18067319 -0.16856100 0.32179502 -0.00048953 0.00046624
      0.00058583 0.00041082 -0.00214614 0.00000982 0.00055434 -0.00073302 -
      0.00217088 0.00010261 0.00050261 -0.00012142 -0.00010990 0.00176918
      0.00278231 0.17960139 -0.10508296 0.09282691 -0.18006973 0.10522409
      0.00019225 0.00081970 -0.00383629 0.00092726 -0.00049355 0.00083611
      0.00162688 -0.00005388 -0.00112037 -0.00000115 0.00010744 0.00039449 -
      0.00268894 0.00136818 0.00099254 -0.16727873 0.09216468 -0.08905820
      0.16722243 -0.09391257 0.09179172
    </matrix>
  </property>

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</property>
<property dictRef="me:sigma" default="true">
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</property>
<property dictRef="me:epsilon" default="true">
  <scalar>50.0</scalar>
</property>
<property dictRef="me:frequenciesScaleFactor" default="true">
  <scalar>1</scalar>
</property>

<property dictRef="me:symmetryNumber" default="true">
  <scalar>1</scalar>
</property>
</propertyList>
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z3="0.020738"/>
    <atom id="a2" elementType="O" x3="0.355551" y3="1.136637"
z3="0.001187"/>
    <atom id="a3" elementType="O" x3="0.706523" y3="-1.096977" z3="-
0.028274"/>
    <atom id="a4" elementType="H" x3="2.183878" y3="0.246463"
z3="0.080632"/>
    <atom id="a5" elementType="H" x3="-0.272242" y3="-1.101711" z3="-
0.079840"/>
    <atom id="a6" elementType="O" x3="-1.886176" y3="-0.147812"
z3="0.068940"/>
    <atom id="a7" elementType="H" x3="-1.875688" y3="0.667496" z3="-
0.460040"/>
  </atomArray>
  <bondArray>
    <bond atomRefs2="a7 a6" order="1"/>
    <bond atomRefs2="a5 a3" order="1"/>
    <bond atomRefs2="a3 a1" order="1"/>
    <bond atomRefs2="a2 a1" order="2"/>
    <bond atomRefs2="a1 a4" order="1"/>
    <bond atomRefs2="a5 a6" order="1"/>
  </bondArray>
  <propertyList>
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    </property>
    <property dictRef="me:vibFreqs">
      <array units="cm-1"></array>
    </property>
  </propertyList>
</molecule>

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<property dictRef="me:spinMultiplicity">
  <scalar units="cm-1">2</scalar>
</property>
<property dictRef="me:hessian">
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0.16081559 0.00148969 -0.16051799 0.17495275 -0.00041655 -0.07913543
0.04508925 0.00389904 0.03766745 -0.14145932 0.04957171 -0.65896285
0.22525464 -0.04540028 0.73063533 -0.29714226 -0.05208773 -0.04218471 -
0.03191326 -0.00517189 -0.07241505 0.51577471 -0.05413598 -0.07592192
0.00047244 -0.00430353 0.02892632 -0.01049748 0.12545890 0.08720530 -
0.02489535 0.00448542 -0.13771969 -0.07432692 -0.01008382 -0.02443379 -
0.06904104 -0.07846524 0.48154481 -0.14226409 -0.03535148 0.11393944
0.01602912 0.00065607 -0.01392798 -0.03303739 -0.01434041 0.02461310
0.16409166 -0.03690310 -0.08036158 0.04663281 -0.00286312 0.01927359
0.00523490 -0.00933155 0.00756675 0.00636115 0.05135378 0.04735158
0.12307486 0.04891254 -0.19474548 0.01058227 0.01131854 -0.04004271 -
0.00855807 -0.00178971 0.00864870 -0.12168543 -0.05836134 0.22477289 -
0.04243875 -0.01693190 0.03807336 0.00727822 0.00233318 0.00338614 -
0.14967110 -0.05097564 0.14226151 -0.00442861 -0.00234169 -0.00265199
0.19137752 -0.01220287 0.00065041 0.00644694 0.00373617 -0.00563628
0.00149519 -0.05651514 -0.04918124 0.07617208 -0.00219729 0.00608387
0.00025169 0.06647960 0.05102427 0.01014964 -0.00048985 0.02711509 -
0.00422366 -0.00111961 -0.01391783 0.17700230 0.08424789 -0.31414507 -
0.00418019 0.00015065 0.00031988 -0.17656786 -0.08129444 0.29501081
0.00684603 0.00055575 0.00404039 0.00008550 0.00012236 -0.00208535 -
0.00240098 -0.00059254 0.00080241 -0.00049930 0.00005751 -0.00068052 -
0.00332531 -0.00093895 -0.00022551 0.31584635 0.00208526 -0.00017155
0.00210907 -0.00236907 0.00105049 -0.00041386 -0.00290692 0.00213780
0.00370128 -0.00022328 -0.00041683 -0.00021030 0.00154634 -0.00471021 -
0.00427695 -0.17880551 0.10719325 -0.01217041 -0.00196317 -0.00925235
0.00230440 0.00068946 0.00588572 0.01356969 0.00608598 -0.01277459
0.00124222 -0.00012562 0.00065223 -0.00181222 -0.00443964 0.00462458
0.16542732 -0.09307392 0.09992261 0.00066709 0.00026291 0.00154117 -
0.00561674 -0.00142221 0.00124692 -0.00160972 -0.00111080 0.00058630
0.00010861 0.00002816 -0.00008111 0.00120802 0.00163849 -0.00195471 -
0.31655228 0.18067319 -0.16856100 0.32179502 -0.00048953 0.00046624
0.00058583 0.00041082 -0.00214614 0.00000982 0.00055434 -0.00073302 -
0.00217088 0.00010261 0.00050261 -0.00012142 -0.00010990 0.00176918
0.00278231 0.17960139 -0.10508296 0.09282691 -0.18006973 0.10522409
0.00019225 0.00081970 -0.00383629 0.00092726 -0.00049355 0.00083611
0.00162688 -0.00005388 -0.00112037 -0.00000115 0.00010744 0.00039449 -
0.00268894 0.00136818 0.00099254 -0.16727873 0.09216468 -0.08905820
0.16722243 -0.09391257 0.09179172
  </matrix>
</property>
<property dictRef="me:sigma" default="true">
  <scalar>5.0</scalar>
</property>
<property dictRef="me:epsilon" default="true">

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    <scalar>50.0</scalar>
  </property>
  <property dictRef="me:frequenciesScaleFactor" default="true">
    <scalar>1</scalar>
  </property>

  <property dictRef="me:symmetryNumber" default="true">
    <scalar>1</scalar>
  </property>
</propertyList>
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0.014726" />
    <atom id="a3" elementType="O" x3="-0.602629" y3="1.0799"
z3="0.041645" />
    <atom id="a4" elementType="H" x3="-2.066" y3="-0.359496"
z3="0.011018" />
    <atom id="a5" elementType="H" x3="0.584288" y3="0.849995"
z3="0.019831" />
    <atom id="a6" elementType="O" x3="1.56661" y3="0.125886" z3="-
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    <atom id="a7" elementType="H" x3="1.78598" y3="-0.261874"
z3="0.761677" />
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    <bond atomRefs2="a6 a7" order="1" />
    <bond atomRefs2="a1 a2" order="2" />
    <bond atomRefs2="a1 a4" order="1" />
    <bond atomRefs2="a1 a3" order="1" id="b1" />
    <bond atomRefs2="a3 a5" order="1" />
    <bond atomRefs2="a3 a6" order="1" id="b2" />
  </bondArray>
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  <property dictRef="me:vibFreqs">
    <array units="cm-1"></array>
  </property>
  <property dictRef="me:spinMultiplicity">
    <scalar units="cm-1">2</scalar>
  </property>
  <property dictRef="me:hessian">
    <matrix matrixType="squareSymmetricLT" rows="21" units="Hartree/Bohr2">

```


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	-0.2618677	0.1697742	0.0035228	0.2748807		
	0.196055	-0.3034398	-0.005743	-0.2157695	0.435756	
	0.0045287	-0.0031882	-0.0669724	-0.003676		
0.0095086	0.0312932					
	-0.1588011	-0.0679177	-0.001339	0.0452037		
0.0051293	-0.000602	0.1963523				
	-0.1195947	-0.4051497	-0.0076535	0.0294575	-	
0.0991302	-0.0028532	0.0663386	0.5692545			
	-0.0026119	-0.0064346	-0.0713954	0.002191	-	
0.0026648	0.0307823	-0.000981	0.012671	0.0521729		
	-0.297524	-0.0561826	-0.0003371	-0.0258641		
0.0069312	-0.0004847	-0.0070284	-0.0054975	-0.0004673		
0.3264265						
	-0.0469902	-0.0481935	-0.0001014	0.0306908	-	
0.0001887	-0.0016511	-0.0435232	-0.0156532	-0.0009387	0.053713	
0.0690862						
	-0.0003783	-0.0003157	-0.05877	0.000366	-0.000706	
0.0158877	-0.0011468	-0.001061	0.0139249	0.0004066		
0.0016179	0.0235453					
	-0.0280778	-0.1277949	-0.0030734	-0.0334547		
0.0377555	0.0024011	0.0453519	0.0335201	-0.0005888		
0.0018985	0.0125955	0.0004506	-0.0261003			
	-0.0386688	0.0484236	0.0009412	0.0130638	-	
0.0243527	-0.0035961	-0.0329061	-0.0585117	-0.0001129	-0.0006641	-
0.0089099	0.0005754	0.0696214	0.0679013			
	-0.0004098	0.0025469	0.0079841	-0.0002829	-	
0.0022529	-0.0058212	-0.0027417	-0.0011179	-0.0383285	0.0004791	-
0.000024	0.0055159	0.0013976	-0.0010716	0.0688528		

		-0.017383	0.0694583	0.0027556	0.0031312	-
0.0286374	0.0019076	-0.1145727	-0.0031862	0.0003626	0.0012389	-
0.0071621	-0.000797	0.0348308	-0.0131562	0.0172588	0.1321723	
		-0.0018249	-0.0256971	-0.0007322	-0.0212085	-
0.0081588	0.0053212	0.0670602	0.0087277	0.0025316	0.0010655	
	0.0022444	-0.0016973	-0.0187182	-0.026028	-0.0149174	-
0.0731397	0.1367152					
		-0.0014348	-0.0106012	0.0014548	-0.0043634	
	0.0029762	-0.0042982	0.0052034	0.0036181	0.0146372	-
0.0006268	0.0010756	-0.0007756	-0.001603	0.0044698	-0.0418221	
	0.0839964	-0.1648366	0.4250678			
		-0.0031905	0.0020879	-0.0009497	-0.0020291	-
0.0014641	-0.0040746	-0.0065057	-0.0010378	0.0020953	0.0008526	
	0.0006761	0.001099	0.0055517	0.00271	-0.0157011	-
0.0394175	0.0467657	-0.0811719	0.0447385			
		0.0004489	-0.0057129	0.0001763	-0.0060083	-
0.0004859	-0.0035413	0.0058189	0.0004626	-0.0050517	0.0006346	
	0.0016146	0.0015866	-0.0069794	0.0014774	0.016837	
	0.0558232	-0.0878034	0.1632982	-0.0497379	0.0904476	
		0.0008854	0.0048801	-0.0003222	0.0022426	-0.001118
	-0.0008713	0.001607	-0.0036035	-0.0017932	0.0010301	
	0.0000216	0.0006718	0.001016	-0.0012058	0.0036189	-0.105484
	0.1743307	-0.394264	0.098703	-0.1733052	0.3929599	

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</property>

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</property>

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</property>

<property dictRef="me:symmetryNumber" default="true">

<scalar>0.5</scalar>

</property>

</propertyList>

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</me:Rotor>

</me:RotorArray>

<me:ExpansionSize>

<property dictRef="size">

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70.5005   -188.1414   -613.996   -282.2113   -267.4196   -159.1857
    10.6358    245.2404    218.9737    347.1025    365.0686    57.3571
    -99.6071   -26.8602   -34.0298   -32.822    101.3592   -45.1851
    113.7221   -3.0028    -13.9042</array>
    </property>
    </me:FourierSeries>
    <me:FourierSeries>
      <property dictRef="coeficients">
        <array units="kJ/mol">0    73.5168    1.63  -3.0783    30.4075    0
        -51.4142    29.9366    -15.2841  -118.2175    0    3.9356
        32.511    50.8838    54.9548    0    -7.425 -20.5373    29.0099
        69.1591    0    21.2285    1.6799    17.794    56.8859</array>
      </property>
      </me:FourierSeries>
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        <property dictRef="coeficients">
          <array units="kJ/mol">0    0    0    0    0    142.7535    -182.2907
          -27.0838    38.7206    57.5148    -37.3965    125.9835    27.7951
          -126.8799    -129.1822    -72.4943    131.8516    -47.6137    72.005
          35.6564    66.3623    -109.8658    1.234 -64.3606    -
69.2083</array>
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          <property dictRef="coeficients">
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            231.0243    13.9756    15.562    0    17.4889    0.0216
            33.0151    -33.4695    0    59.3966    59.1062    -25.4516    -
86.0002    0    83.4031    119.1956    136.3526    112.7835</array>
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        </me:DOSCMMethod>
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z3="0.000006"/>
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z3="0.000010"/>
          <atom id="a3" elementType="O" x3="0.582571" y3="1.208721"
z3="0.000042"/>
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0.000031"/>

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    <atom id="a5" elementType="H" x3="1.537677" y3="1.409716"
z3="0.000072"/>
    <atom id="a6" elementType="O" x3="-2.068018" y3="-0.360218" z3="-
0.000064"/>
    <atom id="a7" elementType="H" x3="-2.183659" y3="0.611532" z3="-
0.000044"/>
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    <bond atomRefs2="a4 a1" order="1"/>
    <bond atomRefs2="a1 a2" order="2"/>
    <bond atomRefs2="a1 a3" order="1" id="b1"/>
    <bond atomRefs2="a3 a5" order="1"/>
    <bond atomRefs2="a6 a1" order="1" id="b2"/>
  </bondArray>
  <propertyList>
    <property dictRef="me:ZPE">
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    </property>
    <property dictRef="me:vibFreqs">
      <array units="cm-1"></array>
    </property>
    <property dictRef="me:spinMultiplicity">
      <scalar units="cm-1">2</scalar>
    </property>
    <property dictRef="me:hessian">
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        -0.2493478  0.6936992

        0.0002668  -0.0003049  0.1505257

        -0.4700424  0.330991    0.000454    0.4814246

        0.3285622  -0.3760176  0.0002682  -0.3880867
0.4448163

        -0.0002127  0.0001355  -0.0577631  -0.0001127  -
0.0001867  0.0240662

        -0.111077  -0.0517919  0.0002937  0.0203135
0.0472442  -0.0001502  0.5641085

        -0.0057866  -0.2606871  0.0003215  0.0305431  -
0.0781663  -0.000194  0.0827322  0.4149408

```

	-0.000022	0.0001149	-0.0511386	-0.0001157	-
0.0001725	0.0241461	-0.0001296	0.0000086	0.0349982	
	0.0110984	0.0016842	-0.0006339	-0.0447471	
0.0079323	0.0004018	-0.0202921	-0.0145723	0.0002995	-
0.0507364					
	-0.025921	-0.0407913	-0.000086	0.0254001	
0.0018779	0.0000834	-0.0270907	-0.0158599	-0.000082	
0.0101159	0.0778352				
	0.0003007	0.0000745	-0.0510996	-0.0003123	
0.0001874	0.0160618	0.0000137	0.0000354	0.0093334	-
0.0000752	0.0000225	0.0258591			
	-0.0105758	-0.0086092	-0.0004439	0.0002465	0.006083
0.0002291	-0.4601707	-0.0977922	0.0001641	0.0034601	-
0.0009535	-0.0000339	0.4678307			
	-0.0575867	-0.0184295	-0.0004367	0.0057421	
0.0060866	0.0002091	-0.0488585	-0.0598052	0.0001575	-
0.0004283	-0.0047736	-0.0000754	0.101275	0.0768845	
	-0.0000678	-0.0000214	0.0059328	-0.0000083	
0.0000419	-0.006049	0.0000261	-0.0000506	-0.0162213	-
0.0001134	0.0000292	0.0040913	-0.0000085	0.0000158	0.0119154
	-0.1133128	-0.0271858	0.0002054	0.0102293	-
0.0000252	-0.0000217	0.0051778	0.0050017	-0.0002171	0.1014054
	0.0032705	-0.0003704	-0.0010833	0.0000395	0.0001417
	0.0269768				
	0.0022634	-0.0019166	0.0006643	-0.0017579	
0.0015559	-0.0001891	-0.0009277	0.001801	-0.0002457	
0.0051026	-0.0159724	-0.0002347	0.000095	0.0000177	-
0.0000148	-0.0453756	0.4956664			
	-0.0002401	0.00001	0.0038138	0.000077	-0.000183
-0.0011887	-0.0000599	-0.0000059	-0.0010535	-0.0000225	-
0.0001665	-0.0044311	0.0000868	0.0001898	0.0007873	0.0003064
0.0000693	0.0038088				
	-0.0065868	0.0042594	-0.0001421	0.0025755	-
0.0017098	-0.0001336	0.0019399	-0.0001258	0.0000207	-0.0001883
	0.0151787	0.0004775	0.0002924	-0.0001831	0.0000301
0.0293932	0.0406002	-0.0001477	0.0313605		
	0.0078165	0.0041429	-0.0004264	-0.0028317	-
0.0001528	0.0001418	-0.0013076	-0.0022233	0.0002193	-0.0098345
0.0023158	-0.0000097	-0.0000981	0.0000195	-0.0000001	0.0642749
0.4811521	0.0002249	-0.0580196	0.4816816		
	-0.0000249	-0.0000086	-0.000271	0.000018	
0.0000446	0.0007268	0.0000062	-0.0001149	-0.0000642	
0.0001436	0.0001994	0.0001851	0.0000063	-0.0000601	-

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0.0004565  -0.0000443  0.0000894  -0.0017366  -0.0001049  -0.0001498
0.0016163
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<property dictRef="me:epsilon" default="true">
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  <scalar>1</scalar>
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</property>
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    <me:Rotor bondRef="b2" periodicity="1">
    </me:Rotor>
  </me:RotorArray>
  <me:ExpansionSize>
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  </property>
  </me:ExpansionSize>
  <me:FourierSeries>
  <property dictRef="coefficients">
    <array units="kJ/mol">2364.5042 -200.942    -1894.9926  35.9313
52.9416    5.3846    -175.1963   543.8824   -159.9105   84.1117
46.789     42.2194   -8.2002    -98.6432   134.8169   -3.504
80.1105    13.0548    5.9817     58.718    70.3189    16.1461
70.0553    8.3763    -4.0272    40.342    80.8994    63.6902
22.8793    44.1731</array>
  </property>
  </me:FourierSeries>
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  <property dictRef="coefficients">
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0.0028     0    -0.1084    22.5687    4.7052    -29.0074
0.0085     0    0.5305     9.5472     2.7247    -10.5098
0.0029     0    1.3228    -6.5430 0.2762    12.3578    -0.0090
1.9753    -19.556    -1.7065    30.8494    -0.0059</array>
  </property>
  </me:FourierSeries>

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    <me:FourierSeries>
    <property dictRef="coefficients">
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      12.2817    2.3446    -31.2369    -1.2493    37.9071    -1.1006
      12.2841    2.3528    -31.2364    -1.2546    37.8984    -0.6812
      7.5961    1.4634    -19.2977    -0.7802    23.4057</array>
    </property>
    </me:FourierSeries>
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    <property dictRef="coefficients">
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      127.3308    -87.2159    20.4957    -0.0016    0    185.793    -
      84.8573    9.2776    -32.7561    -0.0004    0    53.6099    -8.6458
      -8.615    -25.3497    -0.0101    0    37.199    6.7929    -26.6281
      -4.6401    -0.0029</array>
    </property>
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  </me:DOSCMMethod>
</molecule>
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z3="0.020738"/>
    <atom id="a2" elementType="O" x3="0.355551" y3="1.136637"
z3="0.001187"/>
    <atom id="a3" elementType="O" x3="0.706523" y3="-1.096977" z3="-
0.028274"/>
    <atom id="a4" elementType="H" x3="2.183878" y3="0.246463"
z3="0.080632"/>
    <atom id="a5" elementType="H" x3="-0.272242" y3="-1.101711" z3="-
0.079840"/>
    <atom id="a6" elementType="O" x3="-1.886176" y3="-0.147812"
z3="0.068940"/>
    <atom id="a7" elementType="H" x3="-1.875688" y3="0.667496" z3="-
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    <bond atomRefs2="a5 a3" order="1"/>
    <bond atomRefs2="a3 a1" order="1"/>
    <bond atomRefs2="a2 a1" order="2"/>
    <bond atomRefs2="a1 a4" order="1"/>
    <bond atomRefs2="a5 a6" order="1"/>
  </bondArray>
  <propertyList>
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  </property>
  <property dictRef="me:hessian">
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      0.16081559 0.00148969 -0.16051799 0.17495275 -0.00041655 -0.07913543
      0.04508925 0.00389904 0.03766745 -0.14145932 0.04957171 -0.65896285
      0.22525464 -0.04540028 0.73063533 -0.29714226 -0.05208773 -0.04218471 -
      0.03191326 -0.00517189 -0.07241505 0.51577471 -0.05413598 -0.07592192
      0.00047244 -0.00430353 0.02892632 -0.01049748 0.12545890 0.08720530 -
      0.02489535 0.00448542 -0.13771969 -0.07432692 -0.01008382 -0.02443379 -
      0.06904104 -0.07846524 0.48154481 -0.14226409 -0.03535148 0.11393944
      0.01602912 0.00065607 -0.01392798 -0.03303739 -0.01434041 0.02461310
      0.16409166 -0.03690310 -0.08036158 0.04663281 -0.00286312 0.01927359
      0.00523490 -0.00933155 0.00756675 0.00636115 0.05135378 0.04735158
      0.12307486 0.04891254 -0.19474548 0.01058227 0.01131854 -0.04004271 -
      0.00855807 -0.00178971 0.00864870 -0.12168543 -0.05836134 0.22477289 -
      0.04243875 -0.01693190 0.03807336 0.00727822 0.00233318 0.00338614 -
      0.14967110 -0.05097564 0.14226151 -0.00442861 -0.00234169 -0.00265199
      0.19137752 -0.01220287 0.00065041 0.00644694 0.00373617 -0.00563628
      0.00149519 -0.05651514 -0.04918124 0.07617208 -0.00219729 0.00608387
      0.00025169 0.06647960 0.05102427 0.01014964 -0.00048985 0.02711509 -
      0.00422366 -0.00111961 -0.01391783 0.17700230 0.08424789 -0.31414507 -
      0.00418019 0.00015065 0.00031988 -0.17656786 -0.08129444 0.29501081
      0.00684603 0.00055575 0.00404039 0.00008550 0.00012236 -0.00208535 -
      0.00240098 -0.00059254 0.00080241 -0.00049930 0.00005751 -0.00068052 -
      0.00332531 -0.00093895 -0.00022551 0.31584635 0.00208526 -0.00017155
      0.00210907 -0.00236907 0.00105049 -0.00041386 -0.00290692 0.00213780
      0.00370128 -0.00022328 -0.00041683 -0.00021030 0.00154634 -0.00471021 -
      0.00427695 -0.17880551 0.10719325 -0.01217041 -0.00196317 -0.00925235
      0.00230440 0.00068946 0.00588572 0.01356969 0.00608598 -0.01277459
      0.00124222 -0.00012562 0.00065223 -0.00181222 -0.00443964 0.00462458
      0.16542732 -0.09307392 0.09992261 0.00066709 0.00026291 0.00154117 -
      0.00561674 -0.00142221 0.00124692 -0.00160972 -0.00111080 0.00058630
      0.00010861 0.00002816 -0.00008111 0.00120802 0.00163849 -0.00195471 -
      0.31655228 0.18067319 -0.16856100 0.32179502 -0.00048953 0.00046624
      0.00058583 0.00041082 -0.00214614 0.00000982 0.00055434 -0.00073302 -
      0.00217088 0.00010261 0.00050261 -0.00012142 -0.00010990 0.00176918
      0.00278231 0.17960139 -0.10508296 0.09282691 -0.18006973 0.10522409
      0.00019225 0.00081970 -0.00383629 0.00092726 -0.00049355 0.00083611
      0.00162688 -0.00005388 -0.00112037 -0.00000115 0.00010744 0.00039449 -
      0.00268894 0.00136818 0.00099254 -0.16727873 0.09216468 -0.08905820
      0.16722243 -0.09391257 0.09179172
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  <property dictRef="me:sigma" default="true">
    <scalar>5.0</scalar>
  </property>

```



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  <scalar>50.0</scalar>
</property>
<property dictRef="me:frequenciesScaleFactor" default="true">
  <scalar>1</scalar>
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<property dictRef="me:symmetryNumber" default="true">
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</property>
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z3="0.001187"/>
    <atom id="a3" elementType="O" x3="0.706523" y3="-1.096977" z3="-
0.028274"/>
    <atom id="a4" elementType="H" x3="2.183878" y3="0.246463"
z3="0.080632"/>
    <atom id="a5" elementType="H" x3="-0.272242" y3="-1.101711" z3="-
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    <atom id="a6" elementType="O" x3="-1.886176" y3="-0.147812"
z3="0.068940"/>
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  <bondArray>
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    <bond atomRefs2="a5 a3" order="1"/>
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    <bond atomRefs2="a1 a4" order="1"/>
    <bond atomRefs2="a5 a6" order="1"/>
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    0.16081559 0.00148969 -0.16051799 0.17495275 -0.00041655 -0.07913543
    0.04508925 0.00389904 0.03766745 -0.14145932 0.04957171 -0.65896285
    0.22525464 -0.04540028 0.73063533 -0.29714226 -0.05208773 -0.04218471 -
    0.03191326 -0.00517189 -0.07241505 0.51577471 -0.05413598 -0.07592192
    0.00047244 -0.00430353 0.02892632 -0.01049748 0.12545890 0.08720530 -
    0.02489535 0.00448542 -0.13771969 -0.07432692 -0.01008382 -0.02443379 -
    0.06904104 -0.07846524 0.48154481 -0.14226409 -0.03535148 0.11393944
    0.01602912 0.00065607 -0.01392798 -0.03303739 -0.01434041 0.02461310
    0.16409166 -0.03690310 -0.08036158 0.04663281 -0.00286312 0.01927359
    0.00523490 -0.00933155 0.00756675 0.00636115 0.05135378 0.04735158
    0.12307486 0.04891254 -0.19474548 0.01058227 0.01131854 -0.04004271 -
    0.00855807 -0.00178971 0.00864870 -0.12168543 -0.05836134 0.22477289 -
    0.04243875 -0.01693190 0.03807336 0.00727822 0.00233318 0.00338614 -
    0.14967110 -0.05097564 0.14226151 -0.00442861 -0.00234169 -0.00265199
    0.19137752 -0.01220287 0.00065041 0.00644694 0.00373617 -0.00563628
    0.00149519 -0.05651514 -0.04918124 0.07617208 -0.00219729 0.00608387
    0.00025169 0.06647960 0.05102427 0.01014964 -0.00048985 0.02711509 -
    0.00422366 -0.00111961 -0.01391783 0.17700230 0.08424789 -0.31414507 -
    0.00418019 0.00015065 0.00031988 -0.17656786 -0.08129444 0.29501081
    0.00684603 0.00055575 0.00404039 0.00008550 0.00012236 -0.00208535 -
    0.00240098 -0.00059254 0.00080241 -0.00049930 0.00005751 -0.00068052 -
    0.00332531 -0.00093895 -0.00022551 0.31584635 0.00208526 -0.00017155
    0.00210907 -0.00236907 0.00105049 -0.00041386 -0.00290692 0.00213780
    0.00370128 -0.00022328 -0.00041683 -0.00021030 0.00154634 -0.00471021 -
    0.00427695 -0.17880551 0.10719325 -0.01217041 -0.00196317 -0.00925235
    0.00230440 0.00068946 0.00588572 0.01356969 0.00608598 -0.01277459
    0.00124222 -0.00012562 0.00065223 -0.00181222 -0.00443964 0.00462458
    0.16542732 -0.09307392 0.09992261 0.00066709 0.00026291 0.00154117 -
    0.00561674 -0.00142221 0.00124692 -0.00160972 -0.00111080 0.00058630
    0.00010861 0.00002816 -0.00008111 0.00120802 0.00163849 -0.00195471 -
    0.31655228 0.18067319 -0.16856100 0.32179502 -0.00048953 0.00046624
    0.00058583 0.00041082 -0.00214614 0.00000982 0.00055434 -0.00073302 -
    0.00217088 0.00010261 0.00050261 -0.00012142 -0.00010990 0.00176918
    0.00278231 0.17960139 -0.10508296 0.09282691 -0.18006973 0.10522409
    0.00019225 0.00081970 -0.00383629 0.00092726 -0.00049355 0.00083611
    0.00162688 -0.00005388 -0.00112037 -0.00000115 0.00010744 0.00039449 -
    0.00268894 0.00136818 0.00099254 -0.16727873 0.09216468 -0.08905820
    0.16722243 -0.09391257 0.09179172
    </matrix>
  </property>
  <property dictRef="me:sigma" default="true">
    <scalar>5.0</scalar>
  </property>
  <property dictRef="me:epsilon" default="true">
    <scalar>50.0</scalar>
  </property>
  <property dictRef="me:frequenciesScaleFactor" default="true">
    <scalar>1</scalar>

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</property>

<property dictRef="me:symmetryNumber" default="true">
  <scalar>1</scalar>
</property>
</propertyList>
<me:DOSCMMethod default="true" name="QMRotors" />
</molecule>
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  <atomArray>
    <atom id="a1" elementType="C" x3="0.000000" y3="0.408982"
z3="0.000000"/>
    <atom id="a2" elementType="O" x3="-1.170899" y3="0.254212"
z3="0.000000"/>
    <atom id="a3" elementType="O" x3="0.944992" y3="-0.546685"
z3="0.000000"/>
    <atom id="a4" elementType="H" x3="1.807260" y3="-0.114112"
z3="0.000000"/>
  </atomArray>
  <bondArray>
    <bond atomRefs2="a1 a2" order="2"/>
    <bond atomRefs2="a1 a3" order="1" id="b1"/>
    <bond atomRefs2="a3 a4" order="1"/>
  </bondArray>
</propertyList>
<property dictRef="me:ZPE">
  <scalar units="kJ/mol">-81.6609265</scalar>
</property>
<property dictRef="me:vibFreqs">
  <array units="cm-1"></array>
</property>
<property dictRef="me:spinMultiplicity">
  <scalar units="cm-1">2</scalar>
</property>
<property dictRef="me:hessian">
  <matrix matrixType="squareSymmetricLT" rows="12" units="Hartree/Bohr2">
0.82660911 -0.30227171 0.49418563 0. 0. 0.01457088 -0.56571664 0.37330495 0.
0.66880116 0.37475280 -0.36179056 0. -0.39436274 0.31681490 0. 0. -0.01120230
0. 0. 0.00860310 -0.23089482 -0.11070475 0. -0.09868595 0.02678745 0.
0.52442604 -0.05798377 -0.14197119 0. 0.02748986 0.04281186 0. -0.16745690
0.46122585 0. 0. 0.00808875 0. 0. -0.00620308 0. 0. 0.00451231 -0.02999766
0.03967151 0. -0.00439857 -0.00717752 0. -0.19484527 0.19795081 0. 0.22924150
-0.01449732 0.00957612 0. -0.00643207 0.00216381 0. 0.25137420 -0.36206652 0.
-0.23044480 0.35032659 0. 0. -0.01145733 0. 0. 0.00880228 0. 0. -0.00639798 0. 0.
0.00905303
  </matrix>
</property>
<property dictRef="me:sigma" default="true">
  <scalar>5.0</scalar>
</property>

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<property dictRef="me:epsilon" default="true">
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</property>
<property dictRef="me:frequenciesScaleFactor" default="true">
  <scalar>0.9522</scalar>
</property>
<property dictRef="me:symmetryNumber" default="true">
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</property>
</propertyList>
<me:DOSCMMethod default="true" name="QMRotors" />
<me:ExtraDOSCMMethod xsi:type="me: HinderedRotorQM1D">
  <bondRef>b1</bondRef>
  <me:HinderedRotorPotential format="numerical" units="kJ/mol"
expansionSize="10" UseSineTerms="yes">
    <me:PotentialPoint angle=" 0" "potential=" 0 "/>
    <me:PotentialPoint angle=" 10" "potential=" 1.076455 "/>
    <me:PotentialPoint angle=" 20" "potential=" 4.227055 "/>
    <me:PotentialPoint angle=" 30" "potential=" 9.110485 "/>
    <me:PotentialPoint angle=" 40" "potential=" 15.28041 "/>
    <me:PotentialPoint angle=" 50" "potential=" 22.10671 "/>
    <me:PotentialPoint angle=" 60" "potential=" 28.82799 "/>
    <me:PotentialPoint angle=" 70" "potential=" 34.6566 "/>
    <me:PotentialPoint angle=" 80" "potential=" 38.8574 "/>
    <me:PotentialPoint angle=" 90" "potential=" 40.9578 "/>
    <me:PotentialPoint angle=" 100" "potential=" 40.64274 "/>
    <me:PotentialPoint angle=" 110" "potential=" 38.06975 "/>
    <me:PotentialPoint angle=" 120" "potential=" 33.65891 "/>
    <me:PotentialPoint angle=" 130" "potential=" 28.14536 "/>
    <me:PotentialPoint angle=" 140" "potential=" 22.290495 "/>
    <me:PotentialPoint angle=" 150" "potential=" 16.85571 "/>
    <me:PotentialPoint angle=" 160" "potential=" 12.49738 "/>
    <me:PotentialPoint angle=" 170" "potential=" 9.66184 "/>
    <me:PotentialPoint angle=" 180" "potential=" 8.690405 "/>
    <me:PotentialPoint angle=" 190" "potential=" 9.66184 "/>
    <me:PotentialPoint angle=" 200" "potential=" 12.49738 "/>
    <me:PotentialPoint angle=" 210" "potential=" 16.85571 "/>
    <me:PotentialPoint angle=" 220" "potential=" 22.290495 "/>
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    <me:PotentialPoint angle=" 240" "potential=" 33.65891 "/>
    <me:PotentialPoint angle=" 250" "potential=" 38.06975 "/>
    <me:PotentialPoint angle=" 260" "potential=" 40.64274 "/>
    <me:PotentialPoint angle=" 270" "potential=" 40.9578 "/>
    <me:PotentialPoint angle=" 280" "potential=" 38.8574 "/>
    <me:PotentialPoint angle=" 290" "potential=" 34.6566 "/>
    <me:PotentialPoint angle=" 300" "potential=" 28.82799 "/>
    <me:PotentialPoint angle=" 310" "potential=" 22.10671 "/>
    <me:PotentialPoint angle=" 320" "potential=" 15.28041 "/>
    <me:PotentialPoint angle=" 330" "potential=" 9.110485 "/>
    <me:PotentialPoint angle=" 340" "potential=" 4.227055 "/>

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        <me:PotentialPoint angle=" 350 "potential=" 1.076455    "/>
        <me:PotentialPoint angle=" 360 "potential=" 0        "/>
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</me:ExtraDOSCMMethod>
</molecule>
<molecule spinMultiplicity="2" id="water">
    <atomArray>
        <atom id="a1" elementType="H" x3="0.759508" y3="-0.469893"
z3="0.000000"/>
        <atom id="a2" elementType="O" x3="0.000000" y3="0.117473"
z3="0.000000"/>
        <atom id="a3" elementType="H" x3="-0.759508" y3="-0.469893"
z3="0.000000"/>
    </atomArray>
    <bondArray>
        <bond atomRefs2="a1 a2" order="1"/>
        <bond atomRefs2="a2 a3" order="1"/>
    </bondArray>
    <propertyList>
        <property dictRef="me:ZPE">
            <scalar units="kJ/mol">0</scalar>
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            <array units="cm-1"></array>
        </property>
        <property dictRef="me:spinMultiplicity">
            <scalar units="cm-1">1</scalar>
        </property>
        <property dictRef="me:hessian">
            <matrix matrixType="squareSymmetricLT" rows="9" units="Hartree/Bohr2">
0.10503663 -0.21273897 0.43143138 -0.03130607 0.06982192 0.05843716 -
0.10005101 0.20551299 0.05203861 0.13647704 0.20197176 -0.41553118 -
0.11137358 -0.26201210 0.51083023 0.02504278 -0.05778144 -0.06146036
0.06901347 -0.07051948 0.50677778 -0.00498562 0.00722599 -0.02073254 -
0.03642602 0.06004033 -0.09405625 0.04141164 0.01076721 -0.01590020
0.04155165 0.05649911 -0.09529905 0.12830093 -0.06726632 0.11119925
0.00626328 -0.01204048 0.00302320 -0.12105208 0.18189306 -0.44531742
0.11478879 -0.16985258 0.44229422
            </matrix>
        </property>
        <property dictRef="me:sigma" default="true">
            <scalar>5.0</scalar>
        </property>
        <property dictRef="me:epsilon" default="true">
            <scalar>50.0</scalar>
        </property>
        <property dictRef="me:frequenciesScaleFactor" default="true">
            <scalar>1</scalar>
        </property>
    </propertyList>

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```

    <property dictRef="me:symmetryNumber" default="true">
      <scalar>1</scalar>
    </property>
  </propertyList>
  <me:DOSCMMethod default="true" name="QMRotors" />
</molecule>
<molecule spinMultiplicity="2" id="HOCO_OH_TS">
  <atomArray>
    <atom id="a1" elementType="C" x3="-0.766406" y3="-0.083114" z3="-
0.356823"/>
    <atom id="a2" elementType="O" x3="-1.603898" y3="-0.579884"
z3="0.222677"/>
    <atom id="a3" elementType="O" x3="1.191373" y3="0.312290"
z3="0.442063"/>
    <atom id="a4" elementType="H" x3="1.536981" y3="0.720419" z3="-
0.368238"/>
  </atomArray>
  <bondArray>
    <bond atomRefs2="a1 a2" order="2"/>
    <bond atomRefs2="a3 a4" order="1"/>
    <bond atomRefs2="a3 a1" order="1" id="b1"/>
  </bondArray>
  <propertyList>
    <property dictRef="me:ZPE">
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    </property>
    <property dictRef="me:vibFreqs">
      <array units="cm-1"></array>
    </property>
    <property dictRef="me:spinMultiplicity">
      <scalar units="cm-1">2</scalar>
    </property>
    <property dictRef="me:hessian">
      <matrix matrixType="squareSymmetricLT" rows="12" units="Hartree/Bohr2">
0.64111025 0.38311313 0.23287652 -0.45368143 -0.27483783 0.33219897 -
0.65546051 -0.38719857 0.45075185 0.66247553 -0.38628160 -0.23226796
0.27122152 0.38988957 0.23258837 0.44817346 0.27032660 -0.32446733 -
0.45113111 -0.26958410 0.31914344 0.01811574 0.00758629 -0.00485720 -
0.00719725 -0.00423808 0.00582558 0.06043106 0.00314318 0.00154659
0.00219273 -0.00277612 -0.00187725 -0.00011844 0.07417948 0.09141401
0.00763621 0.00652903 -0.00967065 0.00171471 -0.00154527 0.00506274 -
0.15028468 -0.18268469 0.36770902 -0.00376548 -0.00350085 0.00778677
0.00018224 0.00063011 -0.00286793 -0.07134955 -0.07454654 0.14093376
0.07493279 0.00002529 -0.00215515 0.00142358 0.00008512 0.00155684 -
0.00062406 -0.07752769 -0.09108335 0.17770093 0.07741727 0.09168166 -
0.00212825 -0.00201780 0.00193900 -0.00133545 -0.00009215 0.00026115
0.14931630 0.18061040 -0.36310111 -0.14585261 -0.17850045 0.36090096
      </matrix>
    </property>
    <property dictRef="me:sigma" default="true">

```

```

    <scalar>5.0</scalar>
  </property>
  <property dictRef="me:epsilon" default="true">
    <scalar>50.0</scalar>
  </property>
  <property dictRef="me:frequenciesScaleFactor" default="true">
    <scalar>1</scalar>
  </property>

  <property dictRef="me:symmetryNumber" default="true">
    <scalar>1</scalar>
  </property>
</propertyList>
<me:DOSCMMethod default="true" name="QMRotors" />
<me:ExtraDOSCMMethod xsi:type="me:HinderedRotorQM1D">
  <bondRef>b1</bondRef>
  <me:HinderedRotorPotential format="numerical" units="kJ/mol"
expansionSize="10" UseSineTerms="yes">
    <me:PotentialPoint angle=" 0" "potential=" 0      "/>
    <me:PotentialPoint angle=" 10" "potential=" 0.183785  "/>
    <me:PotentialPoint angle=" 20" "potential=" 0.68263    "/>
    <me:PotentialPoint angle=" 30" "potential=" 1.52279    "/>
    <me:PotentialPoint angle=" 40" "potential=" 2.651755   "/>
    <me:PotentialPoint angle=" 50" "potential=" 4.04327    "/>
    <me:PotentialPoint angle=" 60" "potential=" 5.644825   "/>
    <me:PotentialPoint angle=" 70" "potential=" 7.40391    "/>
    <me:PotentialPoint angle=" 80" "potential=" 9.18925    "/>
    <me:PotentialPoint angle=" 90" "potential=" 10.92208   "/>
    <me:PotentialPoint angle=" 100" "potential=" 12.54989   "/>
    <me:PotentialPoint angle=" 110" "potential=" 14.02017   "/>
    <me:PotentialPoint angle=" 120" "potential=" 15.254155  "/>
    <me:PotentialPoint angle=" 130" "potential=" 16.251845  "/>
    <me:PotentialPoint angle=" 140" "potential=" 17.039495  "/>
    <me:PotentialPoint angle=" 150" "potential=" 17.64336   "/>
    <me:PotentialPoint angle=" 160" "potential=" 18.06344   "/>
    <me:PotentialPoint angle=" 170" "potential=" 18.299735  "/>
    <me:PotentialPoint angle=" 180" "potential=" 18.3785    "/>
    <me:PotentialPoint angle=" 190" "potential=" 18.299735  "/>
    <me:PotentialPoint angle=" 200" "potential=" 18.06344   "/>
    <me:PotentialPoint angle=" 210" "potential=" 17.64336   "/>
    <me:PotentialPoint angle=" 220" "potential=" 17.039495  "/>
    <me:PotentialPoint angle=" 230" "potential=" 16.251845  "/>
    <me:PotentialPoint angle=" 240" "potential=" 15.254155  "/>
    <me:PotentialPoint angle=" 250" "potential=" 14.02017   "/>
    <me:PotentialPoint angle=" 260" "potential=" 12.54989   "/>
    <me:PotentialPoint angle=" 270" "potential=" 10.92208   "/>
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    <me:PotentialPoint angle=" 290" "potential=" 7.40391    "/>
    <me:PotentialPoint angle=" 300" "potential=" 5.644825   "/>
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```

        <me:PotentialPoint angle=" 320 "potential=" 2.651755    "/>
        <me:PotentialPoint angle=" 330 "potential=" 1.52279    "/>
        <me:PotentialPoint angle=" 340 "potential=" 0.68263    "/>
        <me:PotentialPoint angle=" 350 "potential=" 0.183785    "/>
        <me:PotentialPoint angle=" 360 "potential=" 0    "/>
    </me:HinderedRotorPotential>
    <me:CalculateInternalRotorInertia phaseDifference ="180.0"/>
</me:ExtraDOSCMMethod>
</molecule>
<molecule spinMultiplicity="2" id="HOCO_H_TS">
    <atomArray>
        <atom id="a1" elementType="C" x3="-0.358592" y3="-2.026237" z3="-
0.104704"/>
        <atom id="a2" elementType="O" x3="-1.138954" y3="-1.981970" z3="-
0.974920"/>
        <atom id="a3" elementType="O" x3="0.721902" y3="-1.936766"
z3="0.434781"/>
        <atom id="a4" elementType="H" x3="1.749054" y3="-1.565997" z3="-
0.313406"/>
    </atomArray>
    <bondArray>
        <bond atomRefs2="a1 a2" order="1"/>
        <bond atomRefs2="a1 a3" order="1" id="b1"/>
        <bond atomRefs2="a3 a4" order="1"/>
    </bondArray>
    <propertyList>
        <property dictRef="me:ZPE">
            <scalar units="kJ/mol">29.5688735</scalar>
        </property>
        <property dictRef="me:vibFreqs">
            <array units="cm-1"></array>
        </property>
        <property dictRef="me:spinMultiplicity">
            <scalar units="cm-1">2</scalar>
        </property>
        <property dictRef="me:hessian">
            <matrix matrixType="squareSymmetricLT" rows="12" units="Hartree/Bohr2">
0.87279189 -0.00946403 0.09697223 0.71169600 -0.00317237 0.72756420 -
0.42311607 0.01954997 -0.41150046 0.47347466 0.03782563 -0.04536651
0.02529194 -0.02027179 0.02275772 -0.49216279 0.00948767 -0.51453124
0.47941782 -0.01843802 0.51851819 -0.29983308 0.03536056 -0.35568142 -
0.07450445 -0.03571634 0.06458844 0.20841650 0.02455795 -0.05496599 -
0.04073165 -0.00785161 0.02491855 0.02930760 -0.06141464 0.02836405 -
0.30800215 -0.03138976 -0.19681444 -0.05839665 0.00521094 -0.04040180
0.40933658 0.03584733 0.24428437 -0.14984274 -0.04544650 0.05548588
0.02414586 0.01816250 -0.05184347 0.16592103 0.04470830 -0.04293779 -
0.04022415 -0.05291955 0.00336026 0.01861207 0.00857344 -0.00230976 -
0.02035725 0.06177042 0.00168338 -0.00966850 -0.01742431 -0.00273388
0.08846893 0.02507446 -0.01621852 -0.00952071 -0.01206486 0.03641485 -
0.11824361 -0.02442328 -0.00706813 0.03929538 0.01141368 -0.01312819
            </matrix>
        </property>
    </propertyList>
</molecule>

```



```

        </matrix>
    </property>
    <property dictRef="me:sigma" default="true">
        <scalar>5.0</scalar>
    </property>
    <property dictRef="me:epsilon" default="true">
        <scalar>50.0</scalar>
    </property>
    <property dictRef="me:frequenciesScaleFactor" default="true">
        <scalar>1</scalar>
    </property>
    <property dictRef="me:symmetryNumber" default="true">
        <scalar>1</scalar>
    </property>
</propertyList>
<me:DOSCMMethod default="true" name="QMRotors" />
<!--><me:ExtraDOSCMMethod xsi:type="me: HinderedRotorQM1D">
    <bondRef>b1</bondRef>
    <me:HinderedRotorPotential format="numerical" units="kJ/mol"
expansionSize="10" UseSineTerms="yes">
        <me:PotentialPoint angle=" 0" "potential=" 0 "/>
        <me:PotentialPoint angle=" 10" "potential=" 0.183785 "/>
        <me:PotentialPoint angle=" 20" "potential=" 0.68263 "/>
        <me:PotentialPoint angle=" 30" "potential=" 1.52279 "/>
        <me:PotentialPoint angle=" 40" "potential=" 2.651755 "/>
        <me:PotentialPoint angle=" 50" "potential=" 4.04327 "/>
        <me:PotentialPoint angle=" 60" "potential=" 5.644825 "/>
        <me:PotentialPoint angle=" 70" "potential=" 7.40391 "/>
        <me:PotentialPoint angle=" 80" "potential=" 9.18925 "/>
        <me:PotentialPoint angle=" 90" "potential=" 10.92208 "/>
        <me:PotentialPoint angle=" 100" "potential=" 12.54989 "/>
        <me:PotentialPoint angle=" 110" "potential=" 14.02017 "/>
        <me:PotentialPoint angle=" 120" "potential=" 15.254155 "/>
        <me:PotentialPoint angle=" 130" "potential=" 16.251845 "/>
        <me:PotentialPoint angle=" 140" "potential=" 17.039495 "/>
        <me:PotentialPoint angle=" 150" "potential=" 17.64336 "/>
        <me:PotentialPoint angle=" 160" "potential=" 18.06344 "/>
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        <me:PotentialPoint angle=" 240" "potential=" 15.254155 "/>
        <me:PotentialPoint angle=" 250" "potential=" 14.02017 "/>
        <me:PotentialPoint angle=" 260" "potential=" 12.54989 "/>
        <me:PotentialPoint angle=" 270" "potential=" 10.92208 "/>
        <me:PotentialPoint angle=" 280" "potential=" 9.18925 "/>
        <me:PotentialPoint angle=" 290" "potential=" 7.40391 "/>

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```

        <me:PotentialPoint angle=" 300 "potential=" 5.644825    "/>
        <me:PotentialPoint angle=" 310 "potential=" 4.04327    "/>
        <me:PotentialPoint angle=" 320 "potential=" 2.651755    "/>
        <me:PotentialPoint angle=" 330 "potential=" 1.52279    "/>
        <me:PotentialPoint angle=" 340 "potential=" 0.68263    "/>
        <me:PotentialPoint angle=" 350 "potential=" 0.183785    "/>
        <me:PotentialPoint angle=" 360 "potential=" 0    "/>
    </me:HinderedRotorPotential>
    <me:CalculateInternalRotorInertia phaseDifference ="0.0"/>
</me:ExtraDOSCMMethod>-->
</molecule>
<molecule spinMultiplicity="2" id="sink_OH">
    <propertyList>
        <property dictRef="me:ZPE">
            <scalar units="kJ/mol">-30</scalar>
        </property>
    </propertyList>
</molecule>
<molecule spinMultiplicity="2" id="sink_H">
    <propertyList>
        <property dictRef="me:ZPE">
            <scalar units="kJ/mol">-70</scalar>
        </property>
    </propertyList>
</molecule>
<molecule spinMultiplicity="2" id="TSadd">
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z3="0.402837"/>
        <atom id="a2" elementType="O" x3="0.192911" y3="1.201359" z3="-
0.218438"/>
        <atom id="a3" elementType="O" x3="-1.288075" y3="-0.552454"
z3="-0.070515"/>
        <atom id="a4" elementType="H" x3="-0.160810" y3="0.064022"
z3="1.493334"/>
        <atom id="a5" elementType="H" x3="-1.359710" y3="-0.348929"
z3="-1.019851"/>
        <atom id="a6" elementType="O" x3="1.225842" y3="-0.720475" z3="-
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      -0.0096416  -0.0061561  0.0043075  0.0084046

      -0.0139154  -0.0154045  0.0125931  0.0085224
    0.0223638

      0.0069051  0.0076918  -0.0151315  -0.0044104  -
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    0.0020139  0.0000181  0.0172648

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      -0.0055188  -0.003121  -0.0114483  0.0007629
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      -0.0121259  -0.00285  -0.005724  0.0026596  -
    0.0004357  0.0020153  -0.0018039  -0.0008533  -0.0081419  0.0550853

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    0.0037723  0.0096713  -0.0016403  0.0008707  -0.0047919  0.0107685
    0.0529692

      -0.0096041  0.0066385  -0.0914178  0.0006849
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0.0052918	0.0003688	-0.0007966	0.0123932	0.0430987		
		-0.0025972	0.0002549	-0.0075225	0.0015092	
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-0.0012461	0.001222	0.0502132	-0.0881839	0.4769838		
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0.0177565						
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0.0049773	0.0015798	-0.0004635	-0.0002413	0.000011	0.002312	-
0.0014533	-0.000406	-0.0002966	0.0002563	0.0001947	0.0141406	
0.0136809						
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0.0012695	0.001077	-0.0001896	-0.0000718	0.0003817	0.003488	-
0.0028806	-0.0003004	-0.0000357	0.0004177	0.0000452	0.0063632	
0.0052645	0.0034847					
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0.0721827	-0.0492	-0.0219979	0.2960425			
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0.0005463	0.0002675	0.0012774	0.0002339	0.0002942	0.0002857	-
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0.0464325	-0.0207017	0.207135	0.1826483			
		-0.000388	0.0018846	0.0006102	-0.0004847	-
0.0000031	-0.0000608	0.0009872	0.0000852	0.0002318	0.0016437	-
0.0005571	-0.0009011	0.0003885	-0.0002017	-0.0001969	-0.023763	-
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</property>

</propertyList>

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      -50.9609 -73.7616 42.9075 43.8356 40.5185 6.288 -
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      13.7986 0 -246.1411 -154.0667 -58.2076 -50.8218 -
      16.8861 0 -4.2228 -4.6144 -8.5627 2.8722 1.1922
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0.12307486 0.04891254 -0.19474548 0.01058227 0.01131854 -0.04004271 -
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 0.00149519 -0.05651514 -0.04918124 0.07617208 -0.00219729 0.00608387
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